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# Paper 1 of 1 — MCB 303 2025

Level 300 • First Semester • Handout

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## SOIL MICROBIOLOGY (MCB 303)

Soil is alive! More life is living below the surface than above in most environments. Soil are put into six groups: earthworms, nematodes, arthropods, bacteria, fungi, and protozoa with each having their own function. They do play an important role in plant, health and water. The cycle of nutrients going thru our environment are mainly driven by these microorganisms. They maintain: Decomposition, Mineralization, Nitrogen cycling, Storage and release of nutrients, Carbon cycling, Take the pollutants out of the water before it reaches underground or surface water etc.

## MICROBIOLOGY OF THE SOIL

Soil is simply defined as the mixture of minerals, organic matter, gases, liquids, and the countless macroscopic and microscopic organisms that together support life on the earth's crust (lithosphere). It is the most abundant ecosystem on earth. Soil develops as a result of the weathering of geological rock formations in the earth's crust. Soil usually consists of mineral particles derived from the eroded rocks and of organic matter derived from the remains of plants, animals, and microorganisms. Soil is a major component of the earth's ecosystem; and soil is a habitat for a wide variety of soil organisms.

The soil is a medium for plant growth; and it offer plants the necessary physical support, air, water, temperature moderation, nutrients, and protection from toxins. Soil also provides readily available nutrients to plants and animals by converting dead organic matter into various nutrient forms. The soil represents a favourable habitat for microorganisms and is inhabited by a wide range of microorganisms, including bacteria, fungi, algae, viruses and protozoa.

The ecological significance of soil microorganisms is limitless; and though these organisms form only a fraction (less than one percent) of the total soil mass, they play important role in supporting plant communities on the earth surface. Many antibiotic-producing microorganisms (inclusive of bacteria and fungi) are mainly found in the soil; and several other microbes (e.g. Nitrosomonas species) that improve the fertility of the soil and aid in the improvement and development of plants or crops are also found in the soil. Soil microorganisms are very important as almost every chemical transformation taking place in the soil involves active contributions from soil microorganisms including bacteria, fungi, protozoa and algae.

Soil microorganisms actively take part in the biogeochemical cycling of nutrients including carbon, sulphur, nitrogen, iron and phosphorus. Nitrogen-fixing bacteria found in the soil transform nitrogen gas present in the soil atmosphere into soluble or usable nitrogenous compounds such as nitrates and ammonium that plant roots can utilize for their optimal growth; and certain soil microorganisms such as the mycorrhizal fungi can also increase the availability of mineral nutrients such as phosphorus to plants.

Soil microbes have a lot to do with maintaining good soil structure, which promotes infiltration and drainage of water, soil aeration, and vigorous root growth and exploration. Soil microbes play an important role in the process of decomposition of organic matter and release of plant nutrients in soil. Soil bacteria, fungi and actinomycetes help to degrade hydrocarbons including waxes, paraffin's, and oils in the soil. Also in the soil are found some pathogenic soil microorganisms that cause disease in plants and thus affects or damage crops. Soil is divided into different segments or horizons based on their content of organic matter, soil microorganisms and the texture of the soil.

Some soil horizons will have high hydraulic conductivities and thus have greater and more rapid fluctuations in soil moisture while some soil horizons will have greater bulk densities with lower effective porosities and thus have lower saturation values. Other soil horizons have clay films that will retain water at field capacity longer than other soil horizons. Soil horizons are distinct layers of soil that form naturally in undisturbed soil over time (Figure 1). The texture of a soil usually refers to the different sizes of mineral components or particles in a given soil sample; and the different layers or parts of soil support different types of microorganisms and macroscopic organisms.

Figure 1. Illustration of the soil profile.

There are different types of soil but clay, loamy and sandy soils are the most common types of soil used for varying agricultural properties.

- **Loamy soil:** Loamy soil is dark in colour and it is soft, dry and crumbly in the hand. It contains a balance of all the three types of soil materials including silt, sand and clay that make up silty soil, sandy soil and clay soil respectively. Loamy soil contains humus – which is lacking in clay, silt and sandy soil. It has a higher pH and calcium levels because of its high content of organic matter. Loamy soil is the best type of soil for agriculture because of its high content of humus.
- **Clay soil:** Clay soils are tinier than sandy soil and they bind easily with water and plant nutrients. Clay soil has a good water retention capacity. The particles found in clay soil are very small and compact; and clay soil absorbs and holds water and creates a drainage problem in the soil. The ability of clay soil to hold water adversely affects the health of plant and its growth. Clay soil has the smallest particles among the three types of soil.
- **Sandy soil:** Sandy soil contains large particles of sand which determine its drainage and aeration capacity. It is the most loose-type of soil. The water and nutrients (particularly nitrogen) quickly drain away from the plant root zone in sandy soil because they have a poor-water-retention capacity. Sandy soil has the largest particles among the different soil types. It is dry and gritty to the touch.
- **Silt soil:** Silt (silty) soil is made up of fine particles. And like clay soil, silt soil holds water but it does not have good aeration around the plant roots. Silty soil has much smaller particles than sandy soil; and so it is smooth to the touch just like sandy soil. Several natural and human factors or activities affect the composition and fertility of

the soil. Some of these factors that have negative impact on the soil include erosion, loss of organic matter, compaction, salinisation, landslides, and contamination from non-biodegradable human wastes.

Factors that affect microbial population in the soil

The population of microorganisms in the soil are affected or influenced by many factors including soil moisture, pH, temperature, aeration, and amounts of organic and inorganic nutrients.

1. Temperature: Each bacterium has an optimum temperature for growth.

Microorganisms have different range of temperature in which they thrive and reproduce. While some organisms can tolerate and thrive in high temperature conditions, other microbes can only thrive in low temperatures. Microorganisms can thus be described as mesophiles, thermophiles or psychrophiles depending on their temperature range.

Mesophiles (mesophilic microbes) grow at temperature range of 25-35°C while thermophiles (thermophilic microbes) grow at higher temperatures such as 45-65°C and above.

Psychrophiles are microbes that thrive at lower temperatures such as 20°C and below. Most microbes are mesophiles and thus soil-borne microorganisms are mesophilic in nature and thus are able to thrive at temperature range of 25-35°C that can be found in the soil.

2. pH: pH denotes the hydrogen ion concentration of any environment. While some microbes can thrive in environments with higher pH levels (alkaline), others manage to thrive in environments with lower pH (acidic). However, it is noteworthy that the optimal growth of most microbes may be inhibited at higher pH levels and lower pH levels. Thus, most microbes including those that live in the soil thrive at neutral pH a level (7.0) that is neither too acidic nor too alkaline in nature.

3. Aeration: Aeration denotes the supply of air or oxygen to a given habitat. The supply of oxygen/air (aeration) in the soil is critical for the optimal growth of aerobic microbes present in the soil. And when the available oxygen in the soil is usurped by aerobes, a condition of anaerobiosis (lack of oxygen) sets in, and this allows anaerobes to thrive in the environment.

4. Soil moisture: Moisture is a critical factor for microbial growth in any environment including the soil. It is required for the growth and multiplication of vegetative microbial cells. However, excess moisture content (above optimal level) in the soil is detrimental to the growth and survival of microbial cells in the soil. This is because, excess water supply in the soil limits the exchange of gases (e.g. oxygen) required for optimal microbial growth.

- Excess water supply in the soil (especially to levels when the soil becomes water-logged) prevents the supply of available oxygen in the soil; and this condition deprives aerobic microbes in the soil the amount of aeration they require for growth and multiplication. Porous (leaky) soils absorb more water than less-porous soil. This makes soils that are porous in nature to easily become waterlogged than the later (i.e.

less porous soils) due to the accumulation of water by the former. Porous soils can easily support the growth of anaerobic microbes than less-porous soils because of the water-logged nature of the former than the later.

5. Amount of organic and inorganic nutrients: Microorganisms require both inorganic and organic nutrient molecules for their optimal growth in any environment including the soil. These microbes can naturally acquire these nutrients from the organic plant and animal matter which they degrade or decompose in the soil. Organic and inorganic nutrients required for microbial growth can also be acquired or sourced from both organic (biofertilizers) and inorganic fertilizers that are used to improve the fertility of the soil. It is however noteworthy that the application of manure (especially inorganic fertilizers) alters the microbial community in the soil; and this adversely affects the microflora of the soil composition. Adequate organic and inorganic matter from the soil is critical for optimal microbial growth in the soil.

- Organic matter Bacterial size is directly related to organic matter. Humus rich localities are richest in bacteria. Addition of carbonaceous materials and ploughing down of green manure or crop residues have a positive effect on the bacterial counts in soil.

- Inorganic nutrients This commonly increase the bacterial numbers but ammonium fertilizers tend to suppress bacterial flora due to the acidity generated through the microbial oxidation of ammonium to nitric acid.

6. Cultivation practices Ploughing and tillage operations increase the number of soil bacteria by improving soil structure and porosity favouring aeration and altering the moisture status and exposing inaccessible organic nutrients to bacterial action.

7. Seasons They play minor role. In temperate regions, a burst of activity occurs in spring when soil becomes warm and rich in organic matter from the previous autumn. Thus the highest count of bacteria are seen in spring and autumn. Bacterial counts decline in hot summer and winter where cells remain in a state of biochemical inactivity.

8. Depth It is a secondary and variable factor. Bacteria are found more in the upper few centimetres. At the very surface, the bacterial community is less due to inadequate moisture and the bactericidal activity of the sun. The highest number is found at 1-2 cm from the surface and therefore the number declines with depth.

#### Soil Bacteria Functional Groups of Soil bacteria

Bacteria perform many important ecosystem services in the soil including improved soil structure and soil aggregation, recycling of soil nutrients, and water recycling. Soil bacteria form microaggregates in the soil by binding soil particles together with their secretions. These microaggregates are like the building blocks for improving soil structure. Improved soil structure increases water infiltration and increases water holding capacity of the soil.

Bacteria perform important functions in the soil, decomposing organic residues from

enzymes released into the soil. Ingham (2009) describes the four major soil bacteria functional groups as decomposers, mutualists, pathogens and lithotrophs. Each functional bacteria group plays a role in recycling soil nutrients

- **Decomposers:** Most consume simple (easy-to-digest) carbon compounds (i.e. they convert energy in soil organic matter into forms useful to the rest of the organisms). A number of decomposers can break down pesticides and pollutants in soil. They are especially important in stopping or retaining nutrients in their cells, thus preventing the loss of nutrients, such as nitrogen, from the rooting zone. Bacteria dominate in tilled soils but they are only 20-30 percent efficient at recycling carbon (C). Bacteria are higher in nitrogen (N) content (10-30 percent nitrogen, 3 to 10 C:N ratio) than most microbes.
- **Mutualists:** This group form partnerships with plants. The most well-known of these are the nitrogen-fixing bacteria. Of the mutualistic bacteria, there are four bacteria types that convert atmospheric nitrogen (N<sub>2</sub>) into nitrogen for plants. There are three types of soil bacteria that fix nitrogen without a plant host and live freely in the soil and these include *Azotobacter*, *Azospirillum* and *Clostridium*.
- **Pathogens** which cause diseases. Pathogenic bacteria cause diseases in plants and a good example are bacteria blights. Healthy and diverse soil bacteria populations produce antibiotics that protect the plants from disease causing organisms and plant pathogens. Diverse bacteria populations compete for the same soil nutrients and water and tend to act as a check and balance system by reducing the disease-causing organism populations. With high microbial diversity, soils have more nonpathogenic bacteria competing with the pathogenic bacteria for nutrients and habitat. *Streptomyces* (actinomycetes) produce more than 50 different antibiotics to protect plants from pathogenic bacteria.
- **Lithotrophs** (literally meaning rock eaters) or **Chemoautotrophs** (which are able to synthesize all of the organic compounds they need from inorganic raw materials in the absence of sunlight.), obtains its energy from compounds of nitrogen, sulfur, iron or hydrogen instead of from carbon compounds. Under well-aerated conditions, sulfur-oxidizing bacteria make the sulfur more plant available while under saturated (anaerobic, low oxygen) soil conditions, sulfur reducing bacteria make sulfur less plant available.

Actinomycetes have large filaments or hyphae and act similar to fungus in processing soil organic residues which are hard to decompose (chitin, lignin, etc.). When farmers plow or till the soil, actinomycetes release “geosmin” as they die which gives freshly turned soil its characteristic smell. Actinomycetes decompose many substances but are more active at high soil pH levels. Actinomycetes are important in forming stable humus, which enhances soil structure, improves nutrient storage, and increases water retention.

Microbial population in the soil is usually large in the rhizosphere region of the soil – where

nutrient materials such as amino acids and vitamins released from the plant root increases the microbial population and their metabolic activities in the soil. When these critical growth nutrients are released by plant tissues, the growth of the plants in the soil is positively influenced and improved upon due to microbial activities and/or metabolism – which release important microbial products that improve the general fertility of the soil. This phenomenon is generally known as the rhizosphere effect of the soil.

#### Microorganisms in the Rhizosphere and Rhizosphere Effect

The rhizosphere region is a highly favorable habitat for the proliferation, activity and metabolism of numerous microorganisms. The rhizosphere microflora can be enumerated intensively by microscopic, cultural and biochemical techniques. Microscopic techniques reveal the types of organisms present and their physical association with the outer root tissue surface / root hairs. The cultural technique most commonly followed is "serial dilution and plate count method" which reveal the quantitative and qualitative population of microflora. At the same time, a cultural method shows the selective enhancement of certain categories of bacteria. The biochemical techniques used are designed to measure a specific change brought about by the plant or by the microflora.

The rhizosphere effect on most commonly found microorganisms viz. bacteria, actinomycetes, fungi, algae and protozoa is as follows:

##### A. Bacteria:

The greater rhizosphere effect is observed with bacteria (R: S values ranging from 10-20 or more) than with actinomycetes and fungi. Gram-negative, rod shaped, non-sporulating bacteria which respond to root exudates are predominant in the rhizosphere (*Pseudomonas*, *Agrobacterium*). While Gram-positive, rods, cocci and aerobic spore forming (*Bacillus*, *Clostridium*) are comparatively rare in the rhizosphere. The most common genera of bacteria are: *Pseudomonas*, *Arthrobacter*, *Agrobacterium*, *Alcaligenes*, *Azotobacter*, *Mycobacterium*, *Flavobacter*, *Cellulomonas*, *Micrococcus* and others have been reported to be either abundant or sparse in the rhizosphere. From the agronomic point of view, the abundance of nitrogen fixing and phosphate solubilizing bacteria in the rhizosphere assumes a great importance. The aerobic bacteria are relatively less in the rhizosphere because of the reduced oxygen levels due to root respiration. The bacterial population in the rhizosphere is enormous in the ranging form  $10^8$  to  $10^9$  per gram of rhizosphere soil. They cover about 4-10% of the total root area occurring profusely on the root hair region and rarely in the root tips. There is predominance of amino acids and growth factors required by bacteria, are readily provided by the root exudates in the region of rhizosphere.

##### B. Fungi:

In contrast to their effects on bacteria, plant roots do not alter / enhance the total count of fungi in the rhizosphere. However, rhizosphere effect is selective and significant on specific fungal genera (*Fusarium*, *Verticillium*, *Aspergillus* and *Penicillium*) which are stimulated. The R:S ratio of fungal population is believed to be narrow in most of the plants, usually not exceeding to 10. The soil / serial dilution and plating technique used for the enumeration of rhizosphere fungi may often give erratic results as most of the spore formers produce

abundant colonies in culture media giving a wrong picture / estimate (eg Aspergilli and Penicillia). In fact, the mycelial forms are more dominant in the field. The zoospore / forming lower fungi such as Phytophthora, Pythium, Aphanomyces are strongly attracted to the roots in response to particular chemical compounds excreted by the roots and cause diseases under favorable conditions. Several fungi eg Gibberella and fujikurio produces phytohormones and influence the plant growth.

#### C. Actinomycetes, Protozoa and Algae:

Stimulation of actinomycetes in the rhizosphere has not been studied in much detail so far. It is generally understood that the actinomycetes are less stimulated in the rhizosphere than bacteria. However, when antagonistic actinomycetes increase in number they suppress bacteria. Actinomycetes may also increase in number when antibacterial agents are sprayed on the crop. Among the actinomycete, the phosphate solublizers (eg. Nocardia, Streptomyces) have a dominant role to play.

As rule actinomycetes, protozoa and algae are not significantly influenced by their proximity to the plant roots and their R: S ratios rarely exceed 2 to 3: 1 and around roots of plants, R: S ratio for these microorganisms may go to high. Because of large bacterial community, an increase in the number or activity of protozoa is expected in the rhizosphere. Flagellates and amoebae are dominant and ciliates are rare in the region.

#### Factors affecting microbial flora of the Rhizosphere / Rhizosphere Effect

The most important factors which affect / influence the microbial flora of the rhizosphere or rhizosphere effect are: soil type & its moisture, soil amendments, soil PH, proximity of root with soil, plant species, and age of plant and root exudates.

A. Soil type and its moisture: In general, microbial activity and population is high in the rhizosphere region of the plants grown in sandy soils and least in the high humus soils, and rhizosphere organisms are more when the soil moisture is low. Thus, the rhizosphere effect is more in the sandy soils with low moisture content.

B. Soil amendments and fertilizers: Soil amendments are anything mixed into topsoil to promote healthy plant growth. They function in a number of ways. For example, they may change the pH of soil or supply nutrients. Fertilizers are primarily valued for their ability to supply nutrients. Soil amendment has emerged as an effective and rapid method for improving soil quality. Soil amendment can introduce carbon (C) sources and other nutrients like nitrogen (N), phosphorous (P), and potassium (K) and maintain soil nutrient homeostasis. Imbalanced use of chemical fertilizers can alter soil pH, and increase pests attack, acidification, and soil crust, which results in a decrease in soil organic carbon and useful organisms, stunting plant growth and yield, and even leading to the emission of greenhouse gases. Manure provides the organic matter to the soil which is prepared from the degradation of waste material of animals and plants. Fertilizer provides the mostly inorganic compound to the soil whose excess amount in soil leads to harmful effects on bacteria as well as on humans.

C. Soil PH/ Rhizosphere PH: Respiration by the rhizosphere microflora may lead to the change in soil rhizosphere PH. If the activity and population of the rhizosphere microflora is

more, then the PH of rhizosphere region is lower than that of surrounding soil or non-rhizosphere soil. Rhizosphere effect for bacteria and protozoa is more in slightly alkaline soil and for that of fungi is more in acidic soils.

D. Proximity of root with Soil: Soil samples taken progressively closer to the root system have increasingly greater population of bacteria, and actinomycetes and decreases with the distance and depth from the root system. Rhizosphere effect decline sharply with increasing distance between plant root and soil.

E. Plant Species: Different plant species inhabit often some what variable microflora in the rhizosphere region. The qualitative and quantitative differences are attributed to variations in the rooting habits, tissue composition and excretion products. In general, legumes show / produce a more pronounced rhizosphere effect than grasses or cereals. Biennials, due to their long growth period exert more prolonged stimulation on rhizosphere effect than annuals.

F. Age of Plant: The age of plant also alter the rhizosphere microflora and the stage of plant maturity controls the magnitude of rhizosphere effect and degree of response to specific microorganisms. The rhizosphere microflora increases in number with the age of the plant and reaching at peak during flowering which is the most active period of plant growth and metabolism. Hence, the rhizosphere effect was found to be more at the time of flowering than in the seedling or full maturity stage of the plants. The fungal flora (especially, Cellulolytic and Amylolytic) of the rhizosphere usually increases even after fruiting and the onset of senescence due to accumulation of moribund tissue and sloughed off root parts / tissues: whereas, bacterial flora of the rhizosphere decreases after the flowering period and fruit setting.

G. Root / exudates /excretion: One of the most important factors responsible for rhizosphere effect is the availability of a great variety of organic substances at the root region by way of root exudates/excretions. The quantitative and qualitative differences in the microflora of the rhizosphere from that of general soil are mainly due to influences of root exudates. The spectrum of chemical composition root exudates varies widely, and hence their influence on the microflora also varies widely.

Root exudates are composed of the chemical substances like:

Root Executes Chemical Substances

Amino Acids All naturally occurring amino acids.

Organic acids Acetic, butyric, citric, fumaric, lactic, malic, propionic, succinic etc.

Arabinose, fructose, galactose, glucose, maltose, mannose, oligosaccharides,

Carbohydrates / sugars

raffinose, ribose, sucrose, xylose etc.

Nucleic acid derivatives Adenine, cystidine, guanine, undine

Growth factors

Biotin, choline, inositol, pyridoxine etc

(phytohormones)

Vitamins Thiamine, nicotinic acid, biotin etc

Enzymes Amylase, invertase, protease, phosphatase etc.

Auxins, glutamine, glycosides, hydrocyanic acid peptides, Uv-absorbing

Other compounds compounds, nematode attracting factors, spore germination stimulators, spore inhibitors etc.

The nature and amount of chemical substances thus exuded are dependent on the species of plant, plant age, inorganic nutrients, and temperature, light intensity, O<sub>2</sub> / CO<sub>2</sub> level, root injury etc. Another source of nutrients for the microorganisms in the rhizosphere region is the sloughed off root epidermis which exert selective stimulation effect on some specific groups of microorganisms. For instance, glucose and amino acids in the exudates readily attract Gram-negative rods which predominantly colonize the roots. Sugars and amino acids in the root exudates stimulate the germination of chlamydospores and other resting spores of fungi; stimulation effect of root exudates on plant pathogenic fungi, nematodes is also well known.

#### Alterations in Rhizosphere Microflora

Foliar application of various chemicals leads to alterations in the rhizosphere microflora by changing the pattern

of root exudates. The pattern of the rhizosphere microflora i.e. numbers and species composition can be changed

/ altered by various factors, such as: (i) Soil amendments, (ii) Foliar application of fertilizers / nutrients,

fungicides, insecticides and hormones and (iii) Bacterization / microbial seed inoculants.

##### A. Soil amendments:

Soil amendments with inorganic and organic fertilizers can alter the rhizosphere microflora and an understanding

of the type of changes in the microflora can be useful in the indirect control of pathogens. Dwivedi and Chaube

(1985) showed that amendment of soil with neem-cake can stimulate the activity of actinomycetes which results

into the reduction of propagules of *Macrophomina phaseolina*. It is also known to control phytopathogenic

nematodes in soil by stimulating nematode trapping fungi. Amendment of soil with castor and bean leaves

stimulate the activity of *Trichoderma viride* and *Penicillium* in the rhizosphere leading to the control of *Sclerotium rolfsii*.

##### B. Foliar application of fertilizers and agrochemicals:

Translocation of photosynthate from leaves to roots takes place as a part of the normal metabolic activity in

plants. Therefore, organic substances, including plant protection chemicals (fungicides, insecticides), growth

regulators and plant nutrients applied to foliage / leaves get absorbed into the leaf tissue and further get

translocated to roots along with photosynthates. Many workers have reported that foliar application with various

chemicals cause marked alterations in the number and kind / qualities of microorganisms in the rhizosphere of

several cereals and leguminous crop plants. Thus, such an approach can be used as a new tool in the biological

control of root diseases, stimulation of activity of nitrogen-fixing bacteria and other beneficial microorganisms

in the soil.

C. Seed treatment with bio inoculants:

Bio inoculants such as Azotobacter, Beijerinckia, Azospirillum, Rhizobium or P -solubilizing microorganisms

(eg. Bacillus, Trichoderma, Azotobacter croococcum, Aspergillus niger, Penicillium digitatum etc.) When applied to

the seed / soil helps in the establishment of beneficial microorganisms in the rhizosphere region which will

further benefit in plant growth, encourage inhibition of plant pathogenic organisms in the root vicinity and enrich

the soil with added microbial bio-mass.

Associative and Antagonistic activities in the Rhizosphere

In natural environments (eg. Soil, Air, Water etc.) a number of relationships exist between individual microbes,

microbial species and between individual cells. The composition of microflora of any habitat (soil / rhizosphere)

is governed by the biological equilibrium created by the associations and interactions of all individuals found in

the community. In soil and rhizosphere region, many microorganisms live in close proximity and their

interactions with each other may be associative or antagonistic.

A. Associative interactions / activities in rhizosphere: The dependence of one microorganism upon another

for extra-cellular products (eg. amino acids & growth promoting substances) can be regarded as an associative

activity / effect in rhizosphere. There is an increase in the exudation of amino acids, organic acids and

monosaccharide by plant roots in the presence of microorganisms. Gibberellins and gibberellin- like substances

are known to be produced by bacterial genera viz Azotobacter, Arthrobacter,

Pseudomonas, and Agrobacterium which are commonly found in the rhizosphere. Microorganisms also

influence root hair development, mucilage secretion and lateral root development. Fungi inhabiting the root

surface facilitate the absorption of nutrient by the roots.

Mycorrhiza is one of the best known associative / symbiotic interactions which exist between the roots of higher

plants and fungi. This mycorrhizal association has been found to improve plant growth through better uptake of

phosphorus and zinc from soil, suppression of root pathogenic fungi and nematodes. Another example is

association between the bacterium *Rhizobium* and roots of legumes and *Azospirillum* with cereal crops (wheat, rye, bajara, maize etc).

B. Antagonistic interactions / activities in rhizosphere: The biochemical qualities of root exudates and the

presence of antagonistic microorganisms, plays important role in encouraging or inhibiting the soil borne plant

pathogens in the rhizosphere region. Several mutualistic, communalistic, competitive and antagonistic

interactions exist in the rhizosphere. The number and qualities of antagonistic microorganisms in the rhizosphere

could be increased through artificial means such as fertilizer application, organic amendments, foliar spraying of

chemicals etc.

Antagonistic microorganisms in the rhizosphere play an important role in controlling some of the soil borne

plant pathogens. Stanier et al (1966) discovered the bacterial strain *Pseudomonas fluorescens* and the fluorescent

pigments of this species in biological control of root pathogens. Strains of *P. fluorescence* are collectively called

as "Fluorescent Pseudomonads". They produce variety of biologically active compounds such as plant growth

substances, cyanides, antibiotics and iron chelating substances called "Siderophores" Rovira and Campbell

(1975), showed that bacterial strains of *P. fluorescens* could lyse the hyphae of *Gaumannomyces graminis* var.

*Tritici*, the causative agent of take-all disease of wheat. Fluorescent pseudomonads (*P. fluorescens*, *P.*

*putida*) are known to produce iron chelating substances called Siderophores. These are low molecular weight,

extra cellular, iron-binding agents produced by pseudomonads in response to low iron stress or when  $Fe^{3+}$  is in

short supply. Thus, iron stress triggers the formation of iron-binding ligands called siderophores. Siderophores

contains the pigments Pyovirdin (Fluorescent) and Pyocyanin (non-Fluorescent) having iron chelating

properties. Another pigment "Pseudobactin" is a fluorescent chelator of iron which is known to promote plant

growth and inhibition of pathogenic bacteria in the rhizosphere. An antibiotic called "Pyrrolnitrin" reduces

damping-off disease in cotton caused by *Rhizoctonia solani*. Several species of *Bacillus* are known to cause

mycolysis in the rhizosphere. eg. *Fusarium oxysporum* hyphae are known to undergo lysis in soil due to these

bacterial metabolites.

The successful antagonists among fungi are *Trichoderma* sp (*T. viride* and *T. harzianum*, *T. hamatum*) and *Gliocladium virens* which parasitize, lyse or kill the phytopathogenic fungi in the soil. Antifungal

and antibacterial actinomycetes in the rhizosphere play an important role in controlling pathogenic fungi and

bacteria, for example *Micromonospora globosa* is a potent antagonist of *Fusarium udum* causing wilt of pigeon

pea. Amoebae are also known to play an antagonistic role in controlling soil fungi, eg. control of take-all disease

of wheat caused by *Gaumannomyces graminis* through the use of *Myxamoebae*. There can also occur

antagonisms between two fungi producing metabolite and interfering the growth of the other fungus as in case

of *Peniophora* antagonizing *Heterobasidium*.

#### Soil Microorganisms in Biodegradation of Pesticides and Herbicides

Pesticides are the chemical substances that kill pests and herbicides are the chemicals that kill weeds. In the context of soil, pests are fungi, bacteria insects, worms, and nematodes etc. that cause damage to field crops. Thus, in broad sense pesticides are insecticides, fungicides, bactericides, herbicides and nematicides that are used to control or inhibit plant diseases and insect pests. Although wide-scale application of pesticides and herbicides is an essential part of augmenting crop yields; excessive use of these chemicals leads to the microbial imbalance, environmental pollution and health hazards. An ideal pesticide should have the ability to destroy target pest quickly and should be able to degrade non-toxic substances as quickly as possible.

The ultimate "sink" of the pesticides applied in agriculture and public health care is soil. Soil being the storehouse of multitudes of microbes, in quantity and quality, receives the chemicals in various forms and acts as a scavenger of harmful substances. The efficiency and the competence to handle the chemicals vary with the soil and its physical, chemical and biological characteristics.

1. Effects of pesticides: Pesticides reaching the soil in significant quantities have direct effect on soil microbiological aspects, which in turn influence plant growth. Some of the

2. most important effects caused by pesticides are : (1) alterations in ecological balance of the soil microflora, (2) continued application of large quantities of pesticides may cause ever lasting changes in the soil microflora, (3) adverse effect on soil fertility and crop productivity, (4) inhibition of N<sub>2</sub> fixing soil microorganisms such as *Rhizobium*, *Azotobacter*, *Azospirillum* etc. and cellulolytic and phosphate solubilizing

microorganisms, (5) suppression of nitrifying bacteria, *Nitrosomonas* and *Nitrobacter* by soil fumigants ethylene bromide, Telone, and vapam have also been reported, (6) alterations in nitrogen balance of the soil, (7) interference with ammonification in soil, (8) adverse effect on mycorrhizal symbioses in plants and nodulation in legumes, and (9) alterations in the rhizosphere microflora, both quantitatively and qualitatively.

2. Persistence of pesticides in soil: How long an insecticide, fungicide, or herbicide persists in soil is of great importance in relation to pest management and environmental pollution. Persistence of pesticides in soil for longer period is undesirable because of the reasons: a) accumulation of the chemicals in soil to highly toxic levels, b) may be assimilated by the plants and get accumulated in edible plant products, c) accumulation in the edible portions of the root crops, d) to be get eroded with soil particles and may enter into the water streams, and finally leading to the soil, water and air pollutions. The effective persistence of pesticides in soil varies from a week to several years depending upon structure and properties of the constituents in the pesticide and availability of moisture in soil. For instance, the highly toxic phosphates do not persist for more than three months while chlorinated hydrocarbon insecticides (eg. DOT, aldrin, chlordane etc) are known to persist at least for 4-5 years and some times more than 15 years.

From the agricultural point of view, longer persistence of pesticides leading to accumulation of residues in soil may result into the increased absorption of such toxic chemicals by plants to the level at which the consumption of plant products may prove deleterious / hazardous to human beings as well as livestock's. There is a chronic problem of agricultural chemicals, having entered in food chain at highly inadmissible levels in India, Pakistan, Bangladesh and several other developing countries in the world. For example, intensive use of DDT to control insect pests and mercurial fungicides to control diseases in agriculture had been known to persist for longer period and thereby got accumulated in the food chain leading to food contamination and health hazards. Therefore, DDT and mercurial fungicides has been, banned to use in agriculture as well as in public health department.

3. Biodegradation of Pesticides in Soil: Pesticides reaching to the soil are acted upon by several physical, chemical, and biological forces. However, physical and chemical forces are acting upon/degrading the pesticides to some extent, microorganism's plays major role in the degradation of pesticides. Many soil microorganisms have the ability to act upon pesticides and convert them into simpler non-toxic compounds. This process of degradation of pesticides and conversion into non-toxic compounds by microorganisms is known as "biodegradation". Not all pesticides reaching to the soil are biodegradable and such chemicals that show complete resistance to biodegradation are called "recalcitrant".

The chemical reactions leading to biodegradation of pesticides fall into several broad categories which are discussed in brief in the following paragraphs.

a) Detoxification: Conversion of the pesticide molecule to a non-toxic compound.

Detoxification is not synonymous with degradation. Since a single change in the side chain of a complex molecule may render the chemical non-toxic.

b) Degradation: The breaking down / transformation of a complex substrate into simpler

products leading finally to mineralization. Degradation is often considered to be synonymous with mineralization, e.g. Thirum (fungicide) is degraded by a strain of *Pseudomonas* and the degradation products are dimethylamine, proteins, sulpholipids, etc.

C. Conjugation (complex formation or addition reaction): In which an organism make the substrate more complex or combines the pesticide with cell metabolites. Conjugation or the formation of addition product is accomplished by those organisms catalyzing the reaction of addition of an amino acid, organic acid or methyl crown to the substrate, for e.g., in the microbial metabolism of sodium dimethyl dithiocarbamate, the organism combines the fungicide with an amino acid molecule normally present in the cell and thereby inactivate the pesticides/chemical.

d) Activation: It is the conversion of non-toxic substrate into a toxic molecule, for eg. Herbicide, 4-butyric acid (2, 4-D B) and the insecticide Phorate are transformed and activated microbiologically in soil to give metabolites that are toxic to weeds and insects.

e) Changing the spectrum of toxicity: Some fungicides/pesticides are designed to control one particular group of organisms / pests, but they are metabolized to yield products inhibitory to entirely dissimilar groups of organisms, for e.g. the fungicide PCNB fungicide is converted in soil to chlorinated benzoic acids that kill plants.

Biodegradation of pesticides / herbicides is greatly influenced by the soil factors like moisture, temperature, PH and organic matter content, in addition to microbial population and pesticide solubility. Optimum temperature, moisture and organic matter in soil provide congenial environment for the break down or retention of any pesticide added in the soil. Most of the organic pesticides degrade within a short period (3-6 months) under tropical conditions. Metabolic activities of bacteria, fungi and actinomycetes have the significant role in the degradation of pesticides.

4. Criteria for Bioremediation / Biodegradation: For successful biodegradation of pesticide in soil, following aspects must be taken into consideration. i) Organisms must have necessary catabolic activity required for degradation of contaminant at fast rate to bring down the concentration of contaminant, ii) the target contaminant must be bioavailability, iii) soil conditions must be congenial for microbial /plant growth and enzymatic activity and iv) cost of bioremediation must be less than other technologies of removal of contaminants.

According to Gales (1952) principal of microbial infallibility, for every naturally occurring organic compound there is a microbe / enzyme system capable its degradation.

5. Strategies for Bioremediation: For the successful biodegradation / bioremediation of a given contaminant following strategies are needed.

a) Passive/ intrinsic Bioremediation: It is the natural bioremediation of contaminant by tile indigenous microorganisms and the rate of degradation is very slow.

b) Biostimulation: Practice of addition of nitrogen and phosphorus to stimulate indigenous microorganisms in soil.

c) Bioventing: Process/way of Biostimulation by which gases stimulants like oxygen and methane are added or forced into soil to stimulate microbial activity.

d) Bioaugmentation: It is the inoculation/introduction of microorganisms in the

contaminated site/soil to facilitate biodegradation.

e) Composting: Piles of contaminated soils are constructed and treated with aerobic thermophilic microorganisms to degrade contaminants. Periodic physical mixing and moistening of piles are done to promote microbial activity.

f) Phytoremediation: Can be achieved directly by planting plants which hyperaccumulate heavy metals or indirectly by plants stimulating microorganisms in the rhizosphere.

g) Bioremediation: Process of detoxification of toxic/unwanted chemicals / contaminants in the soil and other environment by using microorganisms.

h) Mineralization: Complete conversion of an organic contaminant to its inorganic co

Explain biodegradation of pesticides in soil. Discuss five chemical reactions leading to biodegradation of pesticides. criteria for effective biodegradation of pesticides

#### Ecological Association/Interactions among Soil Microorganisms

Microorganisms commonly interacts with each other in a positive or negative manner, neutralism is a rare and perhaps only theoretical phenomenon. Soil is the largest terrestrial ecosystem where a wide variety of relationships exist between different types of soil organisms. The associations existing between different soil microorganisms, whether of a symbiotic or antagonistic nature, influence the activities of microorganisms in the soil. Microflora composition of any habitat is governed by the biological equilibrium created by the associations and interactions of all individuals found in the community. In soil, many microorganisms live in close proximity and interact among them-selves in different ways. Some of the interactions or associations are mutually beneficial, or mutually detrimental or neutral. The various types of possible interactions/associations occurring among the microorganisms in soil can be: a) beneficial i) mutualism ii) commensalisms and iii) proto-cooperation or b) detrimental / harmful - i) ammensalism, ii) antagonism, iii) competition iv) Parasitism and v) predation

#### Neutral associations

Neutral association or neutralism is the association between microorganisms, where two different species of microorganisms occupy the same environment without affecting each other. Such an association might be transitory; as conditions change in the environment, like nutrients availability, there might be a change in the relationship. Neutralism describes the relationship between two species which do interact but do not affect each other.

It is to describe interactions where the fitness of one species has absolutely no effect whatsoever on that of other. True neutralism is extremely unlikely and impossible to prove. When dealing with the complex networks of interactions presented by ecosystems, one cannot assert positively that there is absolutely no competition between or benefit to either species. Since true neutralism is rare or nonexistent, its usage is often extended to situations where interactions are merely insignificant or negligible.

#### Positive associations

a) Beneficial Association/Interactions:

**Mutualism (Symbiosis):** It is a relationship or a type of symbiosis in which both the interacting organisms/partners are benefited from each other. The way/manner in which benefit is derived depends on the type of interactions. When the benefit is in the term of exchange of nutrients, then the relationship is termed as "syntrophism" (Greek meaning: Syn -mutual and trophe = nourishment), for example,

i. Lichen (association of algae or BGA with fungus) in which algae benefits by protection afforded to it by the fungal hyphae from environmental stresses, while the fungus obtains and use CO<sub>2</sub> released by the algae during photosynthesis.

Where the blue green algae are the partners in the lichen association, the heterotrophs (Fungus), benefit from the fixed nitrogen by the blue green algae. The algal or cyanobacterial cells are photosynthetic, and as in plants they reduce atmospheric carbon dioxide into organic carbon sugars to feed both symbionts. Both partners gain water and mineral nutrients mainly from the atmosphere, through rain and dust. The lichen association is a close symbiosis. It extends the ecological range of both partners but is not always obligatory for their growth and reproduction in natural environments, since many of the algal symbionts can live independently.

ii. Another mutualistic association is characterized by different metabolic products from the association as compared with the sum of the products of the separate species.

A mutualism between *Thiobacillus ferrooxidans* and *Beijerinckia lacticogenes* helps in ore leaching. Both of these species when grown in a medium free of added carbon and nitrogen sources and with a sterile ore concentrate, the growth of the two species in association and the resulting effect on the rate of leaching copper from the ore concentrate is observed. Leaching is process of recovery of metal from the ore, where microorganisms play the important role of oxidizing insoluble metal sulphides to soluble sulphates.

iii. Microorganisms may also form mutualistic relationships with plants in soil, an example of which is nitrogen fixing bacteria i.e. *Rhizobium* growing in the roots of legumes. In this *Rhizobium*-legume association, *Rhizobium* bacteria are benefited by protection from the environmental stresses while in turn plant benefited by getting readily available nitrate nitrogen released by the bacterial partner.

iv. The *Anabaena*-*Azolla* is an association between the water fern *Azolla* and the cyanobacterium *Anabaena*. This association is of great importance in paddy fields, where nitrogen is frequently a limiting nutrient.

v. An actinorrhizal symbiosis of actinomycetes, *Frankia* with the roots of *Alnus* and *Casurina* (non-legumes) is common in temperate forest ecosystem for soil nitrogen economy.

vi. Another type of symbiotic association which exists between the roots of higher plants and fungus is Mycorrhiza. In this association fungus gets essential organic nutrients and protection from roots of the plants and allows them to multiply and in turn plants uptake phosphorus, nitrogen and other inorganic nutrients made

available by the fungus.

#### Commensalism

Commensalism refers to a relationship between organisms, in which one species of a pair benefits whereas the other is not affected. This happens commonly in soil with respect to degradation of complex molecules like cellulose and lignin.

- i. For example, many fungi can degrade cellulose to glucose, which is utilized by many bacteria. Many bacteria are unable to utilize cellulose, but they can utilise the fungal breakdown products of cellulose, e.g., glucose and organic acids.
- ii. Another example of commensalism is that of a change in the substrate produced by a combination of species and not by individual species. For example, lignin, which is major constituent of woody plants and is usually resistant to degradation by most of the microorganisms. But in forest soils, lignin is readily degraded by a group of Basidiomycetous fungi and the degraded products are used by several other fungi and bacteria which cannot utilize lignin directly. This type of association is also found in organic matter decomposition process.
- iii. Non-pathogenic *E. coli* in the intestinal tract of humans: *E. coli* is a facultative anaerobe that uses oxygen and lowers the O<sub>2</sub> concentration in the gut which creates a suitable environment for obligate anaerobes such as *Bacteroides*. *E. coli* is a host which remains unaffected by *Bacteroides*.
- iv. *Flavobacterium* (host) and *Legionella pneumophila* (commensal):  
*Flavobacterium* excretes cystine which is used by *Legionella pneumophila* and survives in the aquatic habitat.
- v. Association of *Nitrosomonas* (host) and *Nitrobacter* (commensal) in Nitrification: *Nitrosomonas* oxidize Ammonia into Nitrite and finally, *Nitrobacter* uses nitrite to obtain energy and oxidize it into Nitrate.

#### Proto-cooperation

It is a mutually beneficial association between two species/partners. Proto-cooperation is a form of mutualism, but they do not depend on each other for survival. Unlike symbiosis, proto-cooperation is not obligatory for their existence or performance of a particular activity. In this type of association one organism favors its associate by removing toxic substances from the habitat and simultaneously obtains carbon products made by the associate/partner. Nutritional proto-cooperation between bacteria and fungi has been reported for various vitamins, amino acids and purines in terrestrial ecosystem and are very useful in agriculture. Proto-cooperative associations found beneficial in agriculture are : i) synergism between VAM fungus-legume plants and *Rhizobium* in which nitrogen fixation and phosphorus availability / uptake is much higher resulting in higher crop yields and improved soil fertility, ii) synergism between PSM-legume plants and *Rhizobium* and iii) synergism between plant roots and PGPR in rhizosphere where rhizobacteria restrict the growth of phytopathogens on plant roots and secrete growth promoting substances.

#### Negative associations

##### Antagonism

It is the relationship in which one species of an organism is inhibited or adversely affected by another species in the same environment. The relationship is also known as antagonism. The species, which adversely affect the other is said to be antagonistic. Such organisms may be of great practical importance, since they often produce antibiotics or other inhibitory substances that affect the normal growth processes or survival of other organisms.

Antagonistic relations are most common in nature and are also important for the production of antibiotics. The phenomenon of antagonism may be categorized into three i.e. antibiosis, competition and exploitation.

In the process of antibiosis, the antibiotics or metabolites produced by one organism inhibits another organism. An antibiotic is a microbial inhibitor of biological origin. Innumerable examples of antibiosis are found in soil. For example, *Bacillus* Species from soil produces an antifungal agent, which inhibits growth of several soil fungi. Several species of *Streptomyces* from soil produces antibacterial and antifungal antibiotics. Most of the commercial antibiotics such as streptomycin, chloramphenicol, Terramycin and cyclohexamide have been produced from the mass culture of *Streptomyces*. Thus, species of *Streptomyces* are the largest group of antibiotic producers in soil. Another example of antibiosis is inhibition of *Verticillium* by *Trichoderma*, inhibition of *Rhizoctonia* by a bacterium *Bacillus subtilis*, inhibition of soil fungus *Aspergillus terreus* by a bacterium *Staphylococcus aureus*.

Ammensalism is the interaction between two species, where one species suppresses the growth of other by producing toxins like antibiotics or harmful gases like methane, ethylene, nitrite, or HCN or sulphides and other volatile sulphur compounds. Examples; a.) Lactic acid produced by lactic acid bacteria in the vaginal tract: Lactic acid produced by many normal floras in the vaginal tract is inhibitory to many pathogenic organisms such as *Candida albicans*.

b. Skin normal flora: Fatty acid produced by skin flora inhibits many pathogenic bacteria in the skin

c. *Thiobacillus thiooxidans*: *Thiobacillus thiooxidans* produces sulfuric acid by oxidation of sulfur which is responsible for lowering pH in the culture media which inhibits the growth of most other bacteria

### Competition

Soil is inhabited by different kinds of microorganisms; therefore, they exhibit competition among themselves for nutrients and space. In this kind of situation, the best-adapted microorganism will predominate or eliminate the others, which are dependent upon the same limited nutrient substance. The organisms with inherent ability to grow fast are better competitors. The limiting substrate may result in favoring one species over another. Thus, competition can be defined as "the injurious effect of one organism on another because of the removal of some resource of the environment". This phenomenon can result in major fluctuations in the composition of the microbial population in the soil.

For examples; 1. exogenous nutrients are required for the germination of chlamydospores of *Fusarium*, Oospores of *Aphanomyces* and conidia of *Verticillium dahliae* in soil. But other fungi and soil bacteria deplete these critical nutrients required for spore germination and

thereby hinder the spore germination resulting into the decrease in population. 2. Competition for free space has been, reported to suppress the fungal population by soil bacteria. Therefore, organisms with inherent ability to grow fast are better competitors. 3. Competition between *Paramecium caudatum* and *Paramecium aurelia*: Both species of *Paramecium* feeds on the same bacteria population when these protozoa are placed together. *P. aurelia* grow at a better rate than *P. caudatum* due to competition.

### Parasitism

Parasitism is the relationship between two organisms, in which one organism lives in or on another organism. The parasite is dependent upon the host and feeds on the cells, tissues or fluids of the host organism. The parasite lives in intimate physical contact with the host and forms metabolic association with the host. All major groups of plants, animals, and microorganisms are susceptible to attack by microbial parasites. Parasitism is widely spread in soil communities, for example; bacteriophages (viruses which attack bacteria) are strict intracellular parasites Chytrid fungi, which parasitize algae, as well as other fungi and plants; there are many strains of fungi which are parasitic on algae, plants, animals parasitized by different organisms, earthworms are parasitized by fungi, bacteria, viruses etc.

The bacterial parasite of Gram-negative bacteria *Bdellovibrio bacteriovorus* which is widespread in soil and sewage attaches to a host cell at a special region and eventually causes the lysis of that cell. As a consequence, plaque like areas of lysis appears when these parasites are plated along with their host bacteria. Parasitism is widely spread in soil communities.

Viruses which attack bacteria (bacteriophages), fungi, and algae are strict intracellular parasites since they cannot be cultivated as free-living forms. There are also many strains of fungi which are parasitic on algae and other fungi by penetration into the host. Fungi with antagonistic activity toward plant pathogens have an essential role in plant growth and health. Mycoparasites and presumptive mycoparasites have biocontrol potential, some are responsible for natural suppressiveness of soils to certain plant pathogens. Several species of *Trichoderma* were used successfully against certain pathogenic fungi. *Trichoderma* sp. was used as commercial bio-fungicides to control a range of economically important soil-borne fungal plant pathogens. Soils contain a large number and great diversity of oospore parasites, which may have the potential to reduce populations of plant pathogenic *Phycomycetes* in soil.

**Viruses:** Viruses are obligate intracellular parasites that exhibit great host specificity. There are many viruses that are parasite to bacteria (bacteriophage), fungi, algae, protozoa etc.

**Bdellovibrio:** *Bdellovibrio* is an ectoparasite to many gram-negative bacteria.

### Predation

Predation is an association in which predator organism directly feed on and kills the prey organism. Predators may or may not kill their prey prior to feeding on them, but the act of predation often results in the death of its prey and the eventual absorption of the prey's tissue through consumption.

Many species of the soil-dwelling myxobacteria are predators of other microbes. Many

myxobacteria, e.g., *Myxococcus xanthus*, exhibit several complex social traits, including fruiting body formation and spore formation cooperative swarming with two motility systems, and group predation on both bacteria and fungi. Myxobacteria use gliding motility to search the soil matrix for prey and produce a wide range of antibiotics and lytic compounds that kill and decompose prey cells and break down complex polymers, thereby releasing substrates for growth. The nematophagous fungi are the best predatory soil fungi. Species of *Arthrobotrytis* and *Dactylella* are known as nematode trapping fungi. Protozoan-bacteria in soil: Many protozoans can feed on various bacterial population which helps to maintain the count of soil bacteria at optimum level. *Bdellovibrio*, *Vamparococcus*, *Daptobacter*, etc are examples of predator bacteria that can feed on a wide range of the bacterial population.